

ASHLEY (YE) GAO

🌐 <https://lindagaw.github.io> ✉ ygao18@wm.edu

📍 540 Landrum Dr, Williamsburg, VA 23185

EMPLOYMENT

William & Mary <i>Assistant Professor in Computer Science</i>	<i>2023 - Present</i>
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EDUCATION

University of Virginia Advisor: John A. Stankovic Thesis: <i>Addressing Realisms Faced by Deep Learning Models in Cyber Physical Systems</i> <i>Ph.D. in Computer Science</i>	<i>2019 - 2023</i>
University of Virginia <i>M.S. in Computer Science</i>	<i>2017 - 2019</i>
University of California, San Diego <i>B.S. in Computer Science</i> <i>B.A. in Literatures of the World</i>	<i>2012 - 2017</i>

RESEARCH STATEMENT

My research lies at the intersection of **affective computing**, **audio-language modeling**, and **robust machine learning**. I develop methods for **emotion-aware representation learning and generation**, with a focus on audio large language models, controllable text-to-speech, and speech emotion recognition.

SERVICES

Editorial Roles S5. Associate Editor. Elsevier Smart Health.	<i>2024 - Present</i>
Grant Review Panels (Invited) S4. Panelist. National Institutes of Health (NIH).	<i>2025</i>
S3. Panelist. National Institutes of Health (NIH).	<i>2024</i>
S2. Panelist. National Science Foundation (NSF).	<i>2024</i>
Conference Leadership S1. Demo & Poster Chair. IEEE/ACM CHASE (Connected Health: Applications, Systems and Engineering Technologies).	<i>2024</i>

ACCEPTED PUBLICATIONS

[**ICML'26**] Du, H., Shi, J., Lu, S., Zhou, G., **Gao, Y.**, “Sparse Autoencoders for Interpretable Emotion Control in Text-to-Speech.” Proceedings of the International Conference on Machine Learning (ICML), 2026.

[**InterSpeech'26**] Shi, J., Du, H., Song, X., Hong, Y. A., Zhang, Y., **Gao, Y.**, “Emo-BPO: Emotion Bidirectional Preference Optimization for Diffusion-based Emotional TTS.” InterSpeech, 2026.

[**ACL'26**] Nair, T., Nafee, M. W., Jiang, M., **Gao, Y.**, Chen, H., Zhang, Y., “Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing.” Findings of the Association for Computational Linguistics (ACL), 2026.

- [**ACL'26**] Shi, J., Du, H., Song, X., Hong, Y. A., Zhang, Y., Gao, Y., "AffectCodec: Emotion-Preserving Neural Speech Codec for Expressive Speech Modeling." Findings of the Association for Computational Linguistics (ACL), 2026.
- [**ICASSP'26**] Shi, J., Du, H., He, Y., Hong, Y. A., Gao, Y., "Emotion-Aligned Generation in Diffusion Text-to-Speech Models via Preference-Guided Optimization." IEEE ICASSP, 2026.
- [**ICASSP'26**] Shi, J., Du, H., Hong, Y. A., Gao, Y., "EMO-TTA: Improving Test-Time Adaptation of Audio-Language Models for Speech Emotion Recognition." IEEE ICASSP, 2026.
- [**ICASSP'26**] Shi, J., Du, H., Hong, Y. A., Gao, Y., "Plug-and-Play Emotion Graphs for Compositional Prompting in Zero-Shot Speech Emotion Recognition." IEEE ICASSP, 2026.
- [**ICASSP'26**] Mersa, O., Jiang, M., Gao, Y., Li, Q., Zhang, Y., "Hierarchical Convolution Multibranch Transformer for EEG Signals." IEEE ICASSP, 2026.
- [**ICASSP'26**] Jiang, M., Chen, Y., Liu, G., Sahu, A., Gao, Y., Zhang, Y., "Pairs, Not Labels: Predicting Protein-Phenotype Associations via Link Prediction." IEEE ICASSP, 2026.
- [**EMNLP'25**] Du, H., Shi, J., Myerston, J., Lu, S., Zhou, G., Gao, Y., "Role-Guided Annotation and Prototype-Aligned Representation Learning for Historical Literature Sentiment Classification." Findings of EMNLP, 2025.
- [**InterSpeech'25**] Du, H., Lu, S., Zhou, G., Gao, Y., "Emotion-Aware Audio Large Language Models with Dual Cross-Attention and Context-Aware Instruction Tuning." InterSpeech, 2025.
- [**InterSpeech'25**] Shi, J., Zhang, Y., Gao, Y., "Contrastive Learning for Speech Emotion Domain Generalization via Soft Prompt Tuning." InterSpeech, 2025.
- [**CHASE'25**] Clapham, J., Zhou, M., MacDonald, C., Koltermann, K., Gao, Y., Shao, H., "ElectroMeter: The Practical Electrolyte Measurement System." IEEE/ACM CHASE, 2025.
- [**JMIR'25**] Ko, E., Gao, Y., Wang, P., Wijayasingha, L., Wright, K. D., Gordon, K. C., Wang, H., Stankovic, J. A., & Rose, K. M. "Challenges to recruiting dementia caregiving dyads in community-based settings." JMIR Formative Research, 2025.
- [**TMLR'24**] Macdonald, C., Chu, Z., Stankovic, J., Zhou, G., Shao, H., Gao, Y., "MiddleGAN: Generating Inter-domain Images with GANs." Transactions on Machine Learning Research, 2024.
- [**CHASE'24**] Koltermann, K., Clapham, J., Blackwell, G., Jung, W., Burnet, E., Gao, Y., Shao, H., et al., "Gait-Guard: Turn-aware Freezing of Gait Detection for Non-intrusive Intervention Systems." IEEE/ACM CHASE, 2024. (**Best Paper Award**)
- [**SmartComp'23**] Gao, Y., Baucom, B., Gordon, K., Rose, K., Wang, H., Stankovic, J. "E-ADDA: Unsupervised Adversarial Domain Adaptation Enhanced by a New Mahalanobis Distance Loss." IEEE SmartComp, 2023.
- [**HEALTH'21**] Gao, Y., Salekin, A., Gordon, K., Rose, K., Wang, H., Stankovic, J. "Emotion Recognition Robust to Indoor Environmental Distortions Using Out-of-Distribution Detection." ACM Transactions on Computing for Healthcare, 2021.
- [**PERCOM'21**] Gao, Y., et al. "Out-of-the-Box Deployment to Support Research on In-Home Care of Alzheimer's Patients." IEEE Pervasive Computing, 2021.
- [**JAN'21**] Rose, K. M., et al., Gao, Y., "Smarthealth technology study protocol for dementia caregiving relationships." Journal of Advanced Nursing, 2021.
- [**SENSYS'20**] Gao, Y., et al. "A Monitoring and Recommendation System for In-Home Caregivers (Demo)." ACM SenSys, 2020.

GRANTS

G1. COVA CCI: Agentic AI for Multi-Modal Perception Integrity: Threat Reasoning and Response in Cyber-Physical Transportation Systems. Co-PI. \$39,199. 2026

TEACHING

Graduate-Level

T4. CSCI 680: Affective Computing. W&M. 2026

T3. CSCI 680: Deep Transfer Learning. W&M. 2025

T2. CSCI 516: Fundamentals of Artificial Intelligence and Machine Learning. W&M. 2023, 2024, 2026

Undergraduate-Level

T1. CSCI 416: Fundamentals of Artificial Intelligence and Machine Learning. W&M. 2023, 2024, 2026

THE GALATEA LAB

Mentoring Ph.D. Students

M7. Hyo Jin Jon, Ph.D. Student, William & Mary. 2026 - Present

M6. Siyuan Zhao, Ph.D. Student, William & Mary. 2026 - Present

M5. Jiacheng Shi, Ph.D. Student, William & Mary.
First author publications: ICASSP, ACL, Interspeech 2024 - Present

M4. Hongfei Du, Ph.D. Student, William & Mary.
First author publications: ICML, ACL, EMNLP, Interspeech 2024 - Present

Mentoring Undergraduate Students

M3. Ryan Hong, Undergraduate Student, University of Maryland. 2026 - Present

M2. Meave Meyer, Undergraduate Student, William & Mary. 2026 - Present

M1. Lilly Liu, Undergraduate Student, William & Mary.
Initial placement: Carnegie Mellon University (CMU). 2024 - 2026

Selected Research Directions

D3. Domain-Generalized Audio Sensing — robust speech emotion recognition under distribution shift

D2. Sentiment Steering in LLMs — controllable generation of emotional tone in language models

D1. Audio Stress Detection for ADRD Caregivers — mobile and scalable systems for caregiver support

HONORS AND AWARDS

Research Awards

H7. Best Paper Award, IEEE/ACM CHASE. 2024

H6. Best Paper Runner-Up Award, IEEE SmartComp. 2023

H5. Best Poster Award, Computer Science Symposium, University of Virginia. 2022

Service Awards

H4. Best Service Award, IEEE/ACM CHASE. 2024

Fellowships and Academic Honors

H3. Ph.D. Fellowship, Computer Science, University of Virginia. 2019

H2. Academic Excellence Fellowship, Computer Science, University of Virginia. 2017

H1. Honors with High Distinction, University of California, San Diego. 2016

REVIEWER SERVICE

R6. Primary Reviewer. NeurIPS.	2026
R5. Primary Reviewer. ICML, AAAI, ICLR.	2025
R4. Primary Reviewer. NeurIPS, AAAI, ICLR, ACM Health.	2024
R3. Primary Reviewer. AAAI, NeurIPS, ICLR.	2023
R2. Primary Reviewer. IEEE Transactions on Affective Computing; AAAI; Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp); NeurIPS.	2022
R1. Primary Reviewer. ACM Transactions on Computing for Healthcare.	2021

PROFESSIONAL MEMBERSHIPS

P2. Association for Computing Machinery (ACM). Member.	2025 - Present
P1. Institute of Electrical and Electronics Engineers (IEEE). Member.	2025

PUBLISHED CREATIVE WRITING

Books

C2. <i>The Tempest's Call</i> . Pegasus Publishers, United Kingdom. (To Appear)	2026
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Short Fiction

C1. <i>The Purpose of a Name</i> . Neon Dystopia, United States. [Webpage]	2026
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FEATURED

F1. Honors student of the Department of Literature at UCSD. [Webpage]	2016
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REFERENCES

John A. Stankovic

B. P. America Professor
Department of Computer Science
University of Virginia
85 Engineer's Way, Charlottesville, VA 22903
434-982-2275
jas9f@virginia.edu

Jacobo Myerston

Associate Professor
Department of Literature
University of California, San Diego
UC San Diego Arts and Humanities Building, San Diego, CA 92161
jmyerston@ucsd.edu

Page duBois

Distinguished Professor
Department of Literature
University of California, San Diego
UC San Diego Arts and Humanities Building, San Diego, CA 92161
858-534-3379
pdubois@ucsd.edu